

# ArrayList Warm-Up: What's in the List?

## Reminder: What the Methods Return

| Method                             | What it does                                                                   | What it returns                                       |
|------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------|
| <code>add(E obj)</code>            | Adds <code>obj</code> to the end of the list                                   | <code>boolean</code> (always <code>true</code> )      |
| <code>add(int index, E obj)</code> | Inserts <code>obj</code> at <code>index</code> , shifting later elements right | nothing ( <code>void</code> )                         |
| <code>get(int index)</code>        | Looks at the element at <code>index</code> ; the list is unchanged             | the element at <code>index</code>                     |
| <code>set(int index, E obj)</code> | Replaces the element at <code>index</code> with <code>obj</code>               | the <b>old</b> element that was at <code>index</code> |
| <code>remove(int index)</code>     | Removes the element at <code>index</code> , shifting later elements left       | the element that was removed                          |
| <code>size()</code>                | Counts the elements; the list is unchanged                                     | the number of elements                                |

For each problem, write down the contents of the ArrayList after the code runs, in the same format `System.out.println` would print it (for example, `[3, 5, 7]`). Some problems also ask for the value of a variable.

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## Part 1: No Loops

### Problem 1

```
ArrayList<Integer> nums = new ArrayList<Integer>();
nums.add(4);
nums.add(8);
nums.add(15);
nums.set(1, nums.get(0) + nums.get(2));
nums.add(1, nums.get(2) - 5);
nums.set(3, nums.get(1) * 2);
nums.add(nums.get(0));
```

What is in `nums`?

### Problem 2

```
ArrayList<String> words = new ArrayList<String>();
words.add("red");
words.add("green");
words.add("blue");
String old = words.set(0, "yellow");
words.add(old);
words.set(3, words.set(2, "purple"));
words.add(1, words.get(3));
```

What is in `words`? What is the value of `old`?

### Problem 3

```
ArrayList<Integer> list = new ArrayList<Integer>();
list.add(5);
list.add(10);
list.add(15);
```

```
list.add(20);
list.add(25);
int a = list.remove(2);
list.add(1, a);
list.set(0, list.remove(3));
list.add(list.get(1) - list.get(2));
```

What is in list? What is the value of a?

#### Problem 4

```
ArrayList<String> items = new ArrayList<String>();
items.add("A");
items.add("B");
items.add("C");
items.add("D");
items.add("E");
String x = items.remove(1);
String y = items.set(2, x);
items.add(1, y);
items.set(0, items.remove(items.size() - 1));
```

What is in items? What are the values of x and y?

#### Problem 5

```
ArrayList<Integer> data = new ArrayList<Integer>();
data.add(2);
data.add(4);
data.add(6);
data.add(8);
data.add(10);
data.add(12);
data.set(1, data.remove(4));
data.add(2, data.get(0) + data.remove(3));
data.remove(data.get(0) - 1);
data.set(data.size() - 1, data.get(1) + data.get(2));
```

What is in data?

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## Part 2: With Loops

#### Problem 6

```
ArrayList<Integer> multiples = new ArrayList<Integer>();
for (int i = 1; i <= 10; i++)
{
    if (i % 3 == 0)
    {
        multiples.add(i);
    }
}
```

What is in multiples?

## Problems 7-10

Each of the following problems starts with a fresh copy of this list:

```
ArrayList<Integer> nums = new ArrayList<Integer>();  
nums.add(2);  
nums.add(4);  
nums.add(6);  
nums.add(8);  
nums.add(10);  
nums.add(12);
```

### Problem 7

```
for (int i = 0; i < 3; i++)  
{  
    nums.add(0, i);  
}
```

What is in nums?

### Problem 8

```
for (int i = 0; i < 3; i++)  
{  
    int first = nums.remove(0);  
    nums.add(first + 1);  
}
```

What is in nums?

### Problem 9

```
for (int i = nums.size() - 1; i >= 0; i--)  
{  
    if (nums.get(i) > 5)  
    {  
        nums.remove(i);  
    }  
}
```

What is in nums?

### Problem 10

```
for (int i = 0; i < nums.size(); i++)  
{  
    if (nums.get(i) > 5)  
    {  
        nums.remove(i);  
    }  
}
```

What is in nums? Compare your answer to Problem 9. Why do the two differ?